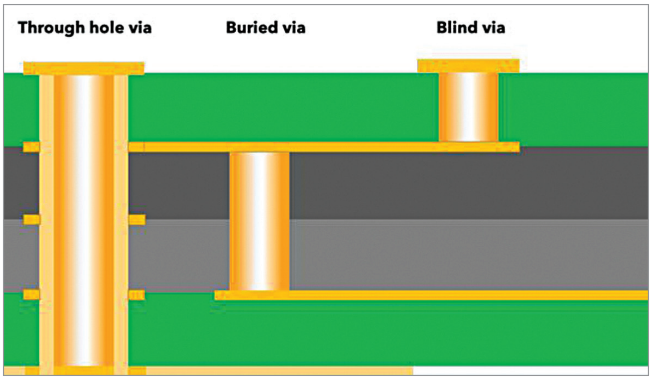


Fig. 3: Types of vias in a PCB (Credit: https://camptechii.com)

SI is a set of measures of the quality of an electrical signal. Over short distances and at low bit rates, a simple conductor can transmit this with sufficient fidelity. At high bit rates and over longer distances, or through various mediums, various effects can degrade the electrical signal to the point where errors occur and the system or device fails.

In designs with high-speed interfaces like DDR3, DDR4, PCIe, etc, pre-layout SI simulations give a rough idea of whether the link will operate with fidelity after the actual layout. Change of positions of DDR and controller may be required to optimise the distance between the source and destination IC. Also, schematic modification may be required to incorporate additional termination circuits.

After the layout is complete, the post-layout SI simulation is performed for final confirmation incorporating all constraints.

## Know your PCB traces

Routing of signals on a PCB is generally done through two types of waveguide structures, namely microstrip and stripline. Stripline is a transmission line trace surrounded by dielectric material suspended between two ground planes on internal layers of a PCB. Microstrip routing is a transmission line trace routed on an external layer of the board. Because of this, it is separated from a single ground plane by a dielectric material.

There are multiple variations of these two types of waveguides available, as shown in Fig. 2. Due to the complexity of the fabrication of PCBs, broadside and asymmetric striplines are seldom used.

## Vias

Via is vertical interconnect access. When a signal travels from one layer to another layer via is used. There are three types of vias in PCBs.

Through-hole vias are those that travel from the top layer to the bottom layer of the PCB. Blind vias are from either the top or bottom layer to an inner layer of the PCB. Buried vias are between two inner layers of the PCB with no access to them after the PCB manufacturing.

Through-hole vias are the most common. Buried and blind vias are costlier to manufacture. Fig. 3 shows different types of vias in a vertical cross-section of a PCB.

## Choosing PCB materials

Due to manufacturability, high-speed PCB layer count is generally an even number. The basic components of PCB manufacturing are core and prepreg.

The core is a layer of dielectric with copper on either side. The layer of dielectric is formed between two smooth foils of copper, to a specified thickness. Prepreg, which is an abbreviation for pre-impregnated, is used by PCB manufacturers to glue together etched cores, or a copper foil to an etched core. Fig. 4 shows the construction details of a 4-layer PCB as an example.

The choice of PCB material depends on the application. FR4 (FR stands for Flame Retardant) material is the most used nowadays. However, very-high-speed designs require other materials like Megtron6.

A few parameters that need to be kept in mind while checking various PCB materials for use are:

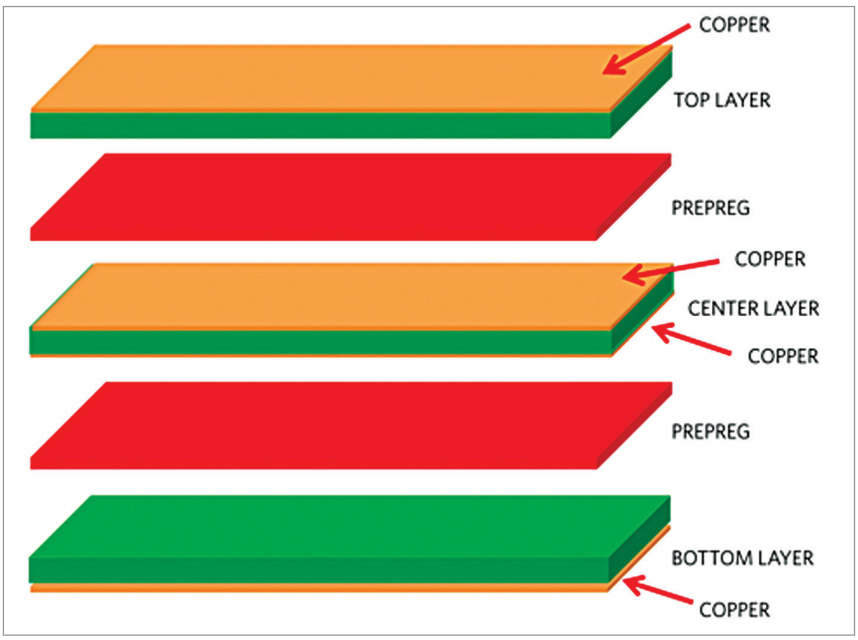


Fig. 4: Construction details of a 4-layer PCB (Credit: https://www.gadgetronicx.com/pcbway-capabilities)